

LIST OF FORMULAE

Arithmetic series

$$T_n = a + (n-1)d \quad S_n = \frac{n}{2}[2a + (n-1)d]$$

Geometric series

$$T_n = ar^{n-1} \quad S_n = \frac{a(r^n - 1)}{r - 1} \quad S_\infty = \frac{a}{1-r}, -1 < r < 1 \text{ or } |r| < 1$$

Circle

$$x^2 + y^2 + 2fx + 2gy + c = 0 \quad (x+f)^2 + (y+g)^2 = r^2$$

Vectors

$$\mathbf{v} = \frac{\mathbf{v}}{|\mathbf{v}|} \quad \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \quad |\mathbf{v}| = \sqrt{x^2 + y^2} \text{ where } \mathbf{v} = x\mathbf{i} + y\mathbf{j}$$

Trigonometry

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

Kinematics

$$v = u + at \quad v^2 = u^2 + 2as \quad s = ut + \frac{1}{2}at^2 \quad v = \frac{dx}{dt} = \dot{x}$$

$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = \ddot{x}$$

Calculus

$$\frac{d}{dx}(ax+b)^n = an(ax+b)^{n-1} \quad \frac{d}{dx} \sin ax = a \cos ax$$

$$\frac{d}{dx} \cos ax = -a \sin ax \quad s = \int v \, dt \quad v = \int a \, dt$$

Statistics

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}, \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^n f_i x_i^2}{\sum_{i=1}^n f_i} - (\bar{x})^2$$

Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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